

NMEC595 - Research Methodology

Lecture 11: Sampling Distribution

(Reference: Research Methodology, C.R. Kothari, 2nd Edition)

Prof. Suparno Bhattacharyya

suparno@iitism.ac.in

Room 224A, Mechanical Engg. Building

Department of Mechanical Engineering
Indian Institute of Technology (ISM) Dhanbad

Winter Semester 2026

Overview

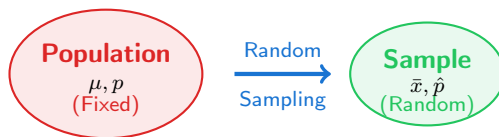
The Central Challenge of Statistics In inferential statistics, we want to use characteristics of the **sample** (a **statistic**) to estimate characteristics of the **population** (a **parameter**).

Population

- Parameters (unknown)
- Mean: μ
- Proportion: p
- Fixed values

Sample

- Statistics (known)
- Mean: $\bar{x}$
- Proportion: $\hat{p}$
- Random variables



Why Mean and Proportion?

In most real problems, we care about **two things**:

The Average (Mean)

What is the typical value?

Example: Average weight of fish in the tank?

The Fraction (Proportion)

What fraction has a property?

Example: Fraction of defective engines?

What is a Sampling Distribution?

Definition:

The sampling distribution of a statistic is a probability distribution based on a large number of samples of size n from a given population.

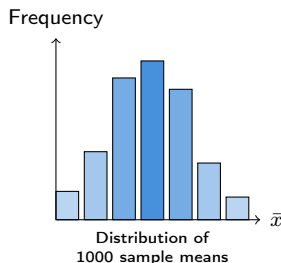
Example: A large tank of fish is being delivered to a lake. We want to know the **average length** of all fish in the tank.

The Process:

- 1 Sample 20 fish $\Rightarrow$ get $\bar{x}_1$
- 2 Sample 20 more $\Rightarrow$ get $\bar{x}_2$
- 3 Repeat 1000 times $\Rightarrow$ get $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_{1000}$

Key Question: Does $\bar{x}_1 = \bar{x}_2$?

Not necessarily! Each sample gives a different result.



Lesson Objectives

What You Will Learn?

Upon completion of this lesson, you should be able to:

- 1 **Understand** the meaning of **sampling distribution**
What happens when we take many samples?
- 2 **Apply** the **Central Limit Theorem** to calculate probabilities
For sample means ($\bar{x}$) and sample proportions ($\hat{p}$)
- 3 **Describe** the sampling distribution characteristics
Shape, center (mean), and spread (standard error)
- 4 **Identify** when to use **Normal** or ***t*-distribution**
Approximation conditions and when they apply

⇒ **Master these to understand how samples help us learn about populations!**

Example

Example: Small Population

The population is the weight of six pumpkins (in pounds) displayed in a carnival “guess the weight” game booth. You are asked to guess the average weight by taking a random sample without replacement from the population.

Pumpkin	A	B	C	D	E	F
Weight (lb)	19	14	15	9	10	17

Since we know the weights from the population, we can find the population mean:

$$\mu = \frac{19 + 14 + 15 + 9 + 10 + 17}{6} = 14 \text{ pounds}$$

Example

Let's obtain all possible samples of size $n = 2$ (without replacement). Since we are drawing at random, each sample will have the same probability of being chosen.

Sample	Weight	$\bar{x}$	Probability
A, B	19, 14	16.5	$\frac{1}{15}$
A, C	19, 15	17.0	$\frac{1}{15}$
A, D	19, 9	14.0	$\frac{1}{15}$
A, E	19, 10	14.5	$\frac{1}{15}$
A, F	19, 17	18.0	$\frac{1}{15}$
B, C	14, 15	14.5	$\frac{1}{15}$
B, D	14, 9	11.5	$\frac{1}{15}$
B, E	14, 10	12.0	$\frac{1}{15}$

Sample	Weight	$\bar{x}$	Probability
B, F	14, 17	15.5	$\frac{1}{15}$
C, D	15, 9	12.0	$\frac{1}{15}$
C, E	15, 10	12.5	$\frac{1}{15}$
C, F	15, 17	16.0	$\frac{1}{15}$
D, E	9, 10	9.5	$\frac{1}{15}$
D, F	9, 17	13.0	$\frac{1}{15}$
E, F	10, 17	13.5	$\frac{1}{15}$

Example

We can combine all of the values and create a table of the possible values and their respective probabilities.

$\bar{x}$	9.5	11.5	12.0	12.5	13.0	13.5	14.0
P	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$
$\bar{x}$	14.5	15.5	16.0	16.5	17.0	18.0	
P	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	

This is the **probability table for the sample mean** and it is the **sampling distribution** of the sample mean weights of the pumpkins when the sample size is 2.

It is also worth noting that the sum of all the probabilities equals 1.

One can see that the chance that the sample mean is exactly the population mean is only 1 in 15, very small.

Example

Now that we have the sampling distribution of the sample mean, we can calculate the mean of all the sample means. In other words, we can find the mean (or expected value) of all the possible $\bar{x}$'s.

The mean of the sample means is:

$$\begin{aligned}
 \mu_{\bar{x}} &= \sum \bar{x}_i f(\bar{x}_i) \\
 &= 9.5 \left(\frac{1}{15} \right) + 11.5 \left(\frac{1}{15} \right) + 12 \left(\frac{2}{15} \right) \\
 &\quad + 12.5 \left(\frac{1}{15} \right) + 13 \left(\frac{1}{15} \right) + 13.5 \left(\frac{1}{15} \right) + 14 \left(\frac{1}{15} \right) \\
 &\quad + 14.5 \left(\frac{2}{15} \right) + 15.5 \left(\frac{1}{15} \right) + 16 \left(\frac{1}{15} \right) + 16.5 \left(\frac{1}{15} \right) \\
 &\quad + 17 \left(\frac{1}{15} \right) + 18 \left(\frac{1}{15} \right) \\
 &= \mathbf{14}
 \end{aligned}$$

Example

Now, let's do the same thing as above but with sample size $n = 5$.

Sample	Weights	$\bar{x}$	Probability
A, B, C, D, E	19, 14, 15, 9, 10	13.4	1/6
A, B, C, D, F	19, 14, 15, 9, 17	14.8	1/6
A, B, C, E, F	19, 14, 15, 10, 17	15.0	1/6
A, B, D, E, F	19, 14, 9, 10, 17	13.8	1/6
A, C, D, E, F	19, 15, 9, 10, 17	14.0	1/6
B, C, D, E, F	14, 15, 9, 10, 17	13.0	1/6

The sampling distribution is:

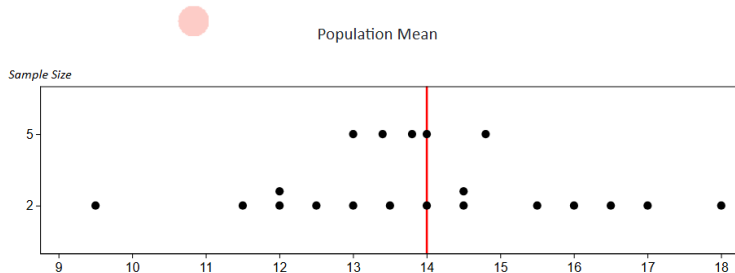
$\bar{x}$	13.0	13.4	13.8	14.0	14.8	15.0
P	1/6	1/6	1/6	1/6	1/6	1/6

The mean of the sample means is:

$$\mu = \left(\frac{1}{6}\right) (13 + 13.4 + 13.8 + 14.0 + 14.8 + 15.0) = 14 \text{ pounds}$$

Distribution of Sampling Error (def. next slide)

The following dot plots show the distribution of the sample means corresponding to sample sizes of $n = 2$ and of $n = 5$.



Sampling Error and Size

Definition:

The error resulting from using a sample characteristic to estimate a population characteristic.

Small Sample ($n = 2$)

- **Wide spread** of $\bar{x}$ values
- **Large** sampling error
- Less precise estimates

Large Sample ($n = 5$)

- **Narrow spread** of $\bar{x}$ values
- **Small** sampling error
- More precise estimates

Key Principle:

Larger sample size $\Rightarrow$ Smaller sampling error $\Rightarrow$ More accurate estimates

Case 1: When Population is Normally Distributed

If the population is normally distributed with mean μ and standard deviation σ , then the sampling distribution of the sample mean is also normally distributed **no matter what the sample size is**.

When the sampling is done with replacement or if the population size is large compared to the sample size, then $\bar{x}$ has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

We use the term **standard error** for the standard deviation of a statistic, and since sample average $\bar{x}$ is a statistic, standard deviation of $\bar{x}$ is also called standard error of $\bar{x}$.

Standard Deviation of $\bar{x}$ [Standard Error]

$$SD(\bar{X}) = SErr(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

Z-Score for Sample Mean

When we know the sample mean is Normal or approximately Normal, then we can calculate a z-score for the sample mean and determine probabilities for it using:

Z-Score of the Sample Mean

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Example

Problem

The engines made by Ford for speedboats have an average power of 220 horsepower (HP) and standard deviation of 15 HP. You can assume the distribution of power follows a normal distribution.

Consumer Reports[®] is testing the engines and will dispute the company's claim if the sample mean is less than 215 HP. If they take a sample of 4 engines, what is the probability the mean is less than 215?

Example

Solution:

We want to find $P(\bar{X} < 215)$.

Since the population follows a normal distribution, we can conclude that $\bar{X}$ has a normal distribution with mean 220 HP ($\mu = 220$) and a standard deviation of $\frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{4}} = 7.5$ HP.

$$P(\bar{X} < 215) = P\left(Z < \frac{215 - 220}{7.5}\right) = P(Z < -0.67) \approx 0.2514$$

If Consumer Reports[®] samples four engines, the probability that the mean is less than 215 HP is **25.14%**.

Try It!

Question

Using the speedboat engines example above, answer the following question:

If Consumer Reports[®] samples 100 engines, what is the probability that the sample mean will be less than 215?

Answer:

The sampling distribution of the sample mean is Normal with mean

$\mu = 220$ and standard deviation $\frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{100}} = 1.5$.

$$P(\bar{X} < 215) = P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} < \frac{215 - 220}{1.5}\right) = P\left(Z < -\frac{10}{3}\right) = 0.00043$$

Comparing Sample Sizes

It is worth noting the difference in the probabilities here.

- When the sample size is $n = 4$, the probability of obtaining a sample mean of 215 or less is **25.14%**.
- When the sample size is $n = 100$, the probability is **0.043%**.

Case 2: Population is Not Normal

What happens when the sample comes from a population that is not normally distributed? This is where the Central Limit Theorem comes in.

Central Limit Theorem

For a large sample size (we will explain this later), $\bar{x}$ is approximately normally distributed, **regardless of the distribution of the population one samples from**. If the population has mean μ and standard deviation σ , then $\bar{x}$ has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The Central Limit Theorem is very powerful! It applies to a sample mean from **any distribution**. We could have a left-skewed or a right-skewed distribution. As long as the sample size is large, the distribution of the sample means will follow an approximate Normal distribution.

For the purposes of this course, a sample size of $n > 30$ is considered a large sample.

Notes on CLT Demonstration

Before we begin the demonstration, let's talk about what we should be looking for:

- If the population is skewed and sample size small, then the sample mean won't be normal.
- When doing a simulation, one replicates the process many times. Using 10,000 replications is a good idea.
- If the population is normal, then the distribution of sample mean looks normal even if $n = 2$.
- If the population is skewed, then the distribution of sample mean looks more and more normal when n gets larger.
- Note that in all cases, the mean of the sample mean is close to the population mean and the standard error of the sample mean is close to $\frac{\sigma}{\sqrt{n}}$.

To work through this demonstration of the central limit theorem yourself, click on the link to the website, Rice Virtual Lab in Statistics > Sampling Distributions, and then click **Begin**.

Sampling Distribution of the Sample Mean

With the Central Limit Theorem, we can finally define the sampling distribution of the sample mean.

Definition 4.3 (Sampling Distribution of the Sample Mean)

The sampling distribution of the sample mean will have:

- The same mean as the population mean, μ .
- Standard deviation [standard error] of $\frac{\sigma}{\sqrt{n}}$.

It will be Normal (or approximately Normal) if either of these conditions is satisfied:

- The population distribution is Normal.
- The sample size is large ($n > 30$).

Example

Problem

The weights of baby giraffes are known to have a mean of 125 pounds and a standard deviation of 15 pounds.

If we obtained a random sample of 40 baby giraffes,

- 1 what is the probability that the sample mean will be between 120 and 130 pounds?
- 2 what is the 75th percentile of the sample means of size $n = 40$?

Example

Solution:

Does the problem indicate that the distribution of weights is normal? No, it does not. In order to apply the Central Limit Theorem, we need a large sample. Since $n = 40 > 30$, we can use the theorem. The sampling distribution of the sample mean is approximately Normal with mean

$\mu = 125$ and standard error $\frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{40}}$.

1. We want $P(120 < \bar{X} < 130)$.

$$\begin{aligned} P(120 < \bar{X} < 130) &= P\left(\frac{120 - 125}{\frac{15}{\sqrt{40}}} < \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} < \frac{130 - 125}{\frac{15}{\sqrt{40}}}\right) \\ &= P(-2.108 < Z < 2.108) \\ &= P(Z < 2.108) - P(Z < -2.108) \\ &= 0.9826 - 0.0174 = 0.9652 \end{aligned}$$

Example

2. To find the 75th percentile, we need the value a such that $P(Z < a) = 0.75$. Using the Z-table or software, we get $a = .6745$.

The formula for the z-score is:

$$z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{40}}} = \frac{\bar{X} - 125}{\frac{15}{\sqrt{40}}}$$

Since we know the z value is 0.6745, we can use algebra to solve for $\bar{X}$:

$$\begin{aligned} 0.6745 &= \frac{\bar{X} - 125}{\frac{15}{\sqrt{40}}} \\ 0.6745 \left(\frac{15}{\sqrt{40}} \right) &= \bar{X} - 125 \\ \bar{X} &= 0.6745 \left(\frac{15}{\sqrt{40}} \right) + 125 \\ &= 126.6 \end{aligned}$$

The 75th percentile of all the sample means of size $n = 40$ is **126.6 pounds**.

Notation for Proportions

Before we begin, let's make sure we review the terms and notation associated with proportions:

- p is the **population proportion**. It is a fixed value.
- n is the size of the random sample.
- $\hat{p}$ is the **sample proportion**. It varies based on the sample.

The following example will illustrate how to find the sampling distribution for an example where the population is small.

Example

Small Population Example

In a particular family, there are five children. Their names are Alex (A), Betina (B), Carly (C), Debbie (D), and Edward (E). The table below shows the child's name and their favorite color.

Name	Alex (A)	Betina (B)	Carly (C)	Debbie (D)	Edward (E)
Color	Green	Blue	Yellow	Purple	Blue

We are interested in the proportion of children in the family who prefer the color blue, and from the table, we can see that $p = 0.40$ of the children prefer blue.

Example

Similar to the pumpkin example earlier in the lesson, let's say we didn't know the proportion of children who like blue as their favorite color. We'll use resampling methods to estimate the proportion. Let's take $n = 2$ repeated samples, taken without replacement. Here are all the possible samples of size $n = 2$ and their respective probabilities of the proportion of children who like blue.

Sample	$\hat{p}$ (Blue)	Probability	Sample	$\hat{p}$ (Blue)	Probability
AB	1/2	1/10	BC	1/2	1/10
AC	0	1/10	BD	1/2	1/10
AD	0	1/10	BE	1	1/10
AE	1/2	1/10	CD	0	1/10
—	—	—	CE	1/2	1/10
—	—	—	DE	1/2	1/10

Example

The probability mass function (PMF) is:

$\hat{p}$ (Blue)	0	1/2	1
Probability	3/10	6/10	1/10

The true proportion is $p = P(\text{Blue}) = \frac{2}{5}$. When the sample size is $n = 2$, you can see from the PMF, it is **not possible to get a sampling proportion that is equal to the true proportion**.

Although not presented in detail here, we could find the sampling distribution for a larger sample size, say $n = 4$. The PMF for $n = 4$ is:

$\hat{p}$ (Blue)	1/4	1/2
Probability	2/5	3/5

Sampling Distribution and Population Proportion

For $n = 2$, the possible values of $\hat{p}$ are 0, $1/2$, and 1, so the true population proportion

$$p = \frac{2}{5}$$

cannot appear as an observed sample proportion.

However, the sampling distribution is still centered at the true population proportion. In fact,

$$E[\hat{p}] = p.$$

Using the PMF for $n = 2$,

$$E[\hat{p}] = 0 \cdot \frac{3}{10} + \frac{1}{2} \cdot \frac{6}{10} + 1 \cdot \frac{1}{10} = \frac{2}{5}.$$

This result is based on **all possible samples** under the sampling process. In practice, we usually observe only a few sample at most, so the observed $\hat{p}$ will vary from sample to sample.

Normal Approximation to the Binomial

For the sampling distribution of the sample mean, we learned how to apply the Central Limit Theorem when the underlying distribution is not normal. In this section, we will present how we can apply the Central Limit Theorem to find the sampling distribution of the sample proportion.

Definition 4.4 (Bernoulli Random Variable Y)

For an experiment that results in a success or a failure, let the random variable Y equal 1, if there is a success, and 0 if there is a failure.

Therefore,

$$Y = \begin{cases} 1 & \text{success} \\ 0 & \text{failure} \end{cases}$$

and let p be the probability of a success.

The Bernoulli random variable is a special case of the Binomial random variable, where the number of trials is equal to one.

Sample Proportion as a Mean

Suppose we have, say n , independent trials of this same experiment. Then we would have n values of Y , namely $Y_1, Y_2, \dots, Y_n$.

If we define X to be the sum of those values, we get:

$$X = \sum_{i=1}^n Y_i$$

X is then a Binomial random variable with parameters n and p .

Suppose we have a random sample of size n from a population and are interested in a particular “success”. Let the probability of success be p . We can label the successes as 1 and the failures as 0. The sample proportion, $\hat{p}$ would be the sum of all the successes divided by the number in our sample. Therefore,

$$\hat{p} = \frac{\sum_{i=1}^n Y_i}{n} = \frac{X}{n}$$

In other words, $\hat{p}$ could be thought of as a mean! If this is the case, we can apply the Central Limit Theorem for large samples!

Mean and Standard Error of $\hat{p}$

Therefore, for large samples, the shape of the sampling distribution for $\hat{p}$ will be approximately normal. What about the mean and the standard deviation?

Mean and Standard Deviation [Standard Error] of $\hat{p}$

Given X is binomial:

The mean of $\hat{p}$:

The mean of $\hat{p}$ would just be p since the mean of X is $\mu = np$ and $\hat{p} = \frac{X}{n}$.

The standard deviation [standard error] of $\hat{p}$:

The standard error of $\hat{p}$ is $\sqrt{\frac{p(1-p)}{n}}$ since the standard deviation of X is $\sqrt{np(1-p)}$.

Conditions for Normal Approximation

The distribution of the sample proportion approximates a normal distribution under the following 2 conditions.

Over the years the values of the conditions have changed. The examples that follow in the remaining lessons will use the first set of conditions at 5, however, you may come across other books or software that may use 10 or 15 for this value.

Source	Condition 1	Condition 2
Book (R)	$np \geq 5$	$n(1 - p) \geq 5$
1990–2000s	$np \geq 10$	$n(1 - p) \geq 10$
Current	$np \geq 15$	$n(1 - p) \geq 15$

Sampling Error for Proportions

Example 4.6 (Resampling with StatKey):

Using **StatKey**, we resample 1000 times from populations that have probabilities of success, 0.1, 0.9, and 0.5 respectively with a sample size of $n = 25$.

Mean and Standard Error of $\hat{p}$

Mean and Standard Error of $\hat{p}$

Let

$$X \sim \text{Binomial}(n, p), \quad \hat{p} = \frac{X}{n}.$$

Then

$$E[\hat{p}] = E\left[\frac{X}{n}\right] = \frac{1}{n}E[X] = \frac{1}{n}(np) = p$$

and

Mean and Standard Error of $\hat{p}$

Let

$$X \sim \text{Binomial}(n, p), \quad \hat{p} = \frac{X}{n}.$$

Then

$$E[\hat{p}] = E\left[\frac{X}{n}\right] = \frac{1}{n}E[X] = \frac{1}{n}(np) = p$$

and

$$\text{SE}(\hat{p}) = \sqrt{\text{Var}(\hat{p})} = \sqrt{\text{Var}\left(\frac{X}{n}\right)} = \sqrt{\frac{1}{n^2}\text{Var}(X)} = \sqrt{\frac{p(1-p)}{n}}.$$

So, the sampling distribution of $\hat{p}$ is centered at p and gets less spread out as n increases.

Sampling Distribution of the Sample Proportion

Definition 4.5

If any set of the two conditions listed above are satisfied, the **sampling distribution of the sample proportion** is:

- approximately normal with mean, $\mu = p$
- standard deviation [standard error], $\sigma = \sqrt{\frac{p(1-p)}{n}}$

If the sampling distribution of $\hat{p}$ is approximately normal, we can convert a sample proportion to a z-score using the following formula:

Z-Score for Sample Proportion

$$z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

Example

Problem:

Suppose it is known that 43% of Americans own an iPhone. If a random sample of 50 Americans were surveyed, what is the probability that the proportion of the sample who owned an iPhone is between 45% and 50%?

Solution:

For this problem, we know $p = 0.43$ and $n = 50$. First, we should check our conditions for the sampling distribution of the sample proportion.

$np = 50(0.43) = 21.5$ and $n(1 - p) = 50(1 - 0.43) = 28.5$ – both are greater than 5.

Example

Since the conditions are satisfied, $\hat{p}$ will have a sampling distribution that is approximately normal with mean $\mu = 0.43$ and standard deviation

[standard error] $\sqrt{\frac{0.43(1-0.43)}{50}} \approx 0.07$.

$$\begin{aligned} P(0.45 < \hat{p} < 0.5) &= P\left(\frac{0.45 - 0.43}{0.07} < \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} < \frac{0.5 - 0.43}{0.07}\right) \\ &= P(0.286 < Z < 1) \\ &= P(Z < 1) - P(Z < 0.286) \\ &= 0.8413 - 0.6126 = 0.2287 \end{aligned}$$

Therefore, if the true proportion of Americans who own an iPhone is 43%, then there would be a **22.87% chance** that we would see a sample proportion between 45% and 50% when the sample size is 50.

Try It!

Question: If a random sample of size of 75 was surveyed, what is the probability we would find more than 50% of Americans with an iPhone?

Try It!

Question: If a random sample of size of 75 was surveyed, what is the probability we would find more than 50% of Americans with an iPhone?

Solution:

First, check our conditions: $np = 75(0.43)$ and $n(1 - p) = 75(1 - 0.43)$ are both greater than five. The sampling distribution of the sample proportion is approximately Normal with:

Mean: $\mu = 0.43$

Standard deviation: $\sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.43(1-0.43)}{75}} \approx 0.05717$

$$P(\hat{p} > 0.5) = P\left(\frac{\hat{p} - 0.43}{\sqrt{\frac{0.43(1-0.43)}{75}}} > \frac{0.5 - 0.43}{\sqrt{\frac{0.43(1-0.43)}{75}}}\right)$$

$$\approx P(Z > 1.22) = 1 - P(Z < 1.22) = 1 - 0.8888 = 0.1112$$

Therefore, there is a **11.1% chance** to get a sample proportion of 50% or higher in a sample size of 75.

Try it!

Question: Suppose it is known that 4% of adults have a certain rare blood type. If a random sample of 30 adults is selected, what is the probability that the proportion of the sample with this blood type is between 0% and 10%?

Try it!

Question: Suppose it is known that 4% of adults have a certain rare blood type. If a random sample of 30 adults is selected, what is the probability that the proportion of the sample with this blood type is between 0% and 10%?

Here,

$$np = 30(0.04) = 1.2, \quad n(1 - p) = 30(0.96) = 28.8.$$

Since $np < 10$, the success-failure condition is not satisfied, so the sampling distribution of $\hat{p}$ is not well approximated by a normal distribution.

Lesson Summary

In this Lesson, we learned how to use the Central Limit Theorem to find the sampling distribution for the sample mean and the sample proportion under certain conditions.

We learned that the sampling distributions are centered around the population parameter with variability. All of this theory was built knowing the parameter. Can we use this information in situations where the parameter is unknown?

The answer is yes! In the next lesson, we take our first step into inference and into the “real world” where the population parameters are unknown and need to be estimated.